Movement of a Charged Particle around a Dipole

Computational Science

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1 Abstract

This project presents a simulation of the motion of a particle around a dipole simulated using Python and numerical methods. The system is modeled according to physical laws such as Coulomb's Law and Newton's Second Law of Motion. Based on these laws, I derived a set of differential equations, which is then used to simulate the particle's trajectory via numerical methods. I chose several initial conditions that result in interesting trajectories using Verlet's integration method. I then conducted a comparative energy conservation analysis between Euler's method and Velocity Verlet integration which confirmed that Verlet's method is more effective at conserving the total mechanical energy of the system. In addition, the system was tested for chaotic properties by introducing microscopic changes of 10^{-6} to the initial position and velocity of the particle and assessing the divergence of the trajectories. Log-distance separation plots showed that these minor changes lead to divergence over time, demonstrating that the dipole environment is highly sensitive to initial conditions. Overall this project shows the utility of numerical methods to understand the properties and behavior of complex physical systems.

2 Introduction

Charged particles are the fundamental building blocks of the matter that surrounds us. What makes charged particles different from other objects is that they are affected by electromagnetic forces. These electromagnetic forces consist of an electric component, created by charge, and a magnetic component, produced by the motion of that charge and the existence of a magnetic field. One example of an electric force acting on a charged particle is when the particle is placed near a dipole. A dipole is a system that includes two fixed, equal and opposite charges separated by a fixed distance (Dipole System).

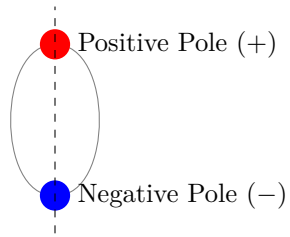


Figure 1: Schematic of the electrostatic dipole system.

In order to compute the movement of a charged particle around a dipole, a combination of Coulomb's Law and Newton's Second Law of Motion can be used to derive equations that describe the particle's acceleration as a vector. This involves solving a set of differential equations. However, this system has no general closed-form solution for arbitrary initial conditions. Instead of an analytical solution, numerical methods typically use a computational simulation to approximate the solution to a complex mathematical problem. Therefore, numerical methods may provide a solution that approximates the movement of the charged particle.

In this project I will use two different numerical methods for solving differential equations, Euler's method and Verlet integration. Utilizing these methods, I will investigate 3 dimensional trajectories of charged particles around a dipole with varying properties such as mass, amplitude and sign of charge, starting position relative to the dipole and starting velocity. I will explore the stability of the two numerical methods in terms of energy conservation, simulate special trajectories and assess wether the system is chaotic, i.e., sensitive to starting properties.

3 Theoretical Basis

3.1 Governing Physical Laws

3.1.1 Particle Movement

To understand how a charged particle moves in the surroundings of a dipole, we must look at the basic laws of physics that govern its behavior. The most important factors are the electrostatic forces created by the two fixed poles and the fundamental laws of motion that determine how the particle responds to these forces. By calculating these interactions, we can predict the particle's path and track how its total energy changes over time.

According to Coulomb's Law the electric force acting on a charged particle is proportional to the magnitude of the charges of the particles and inversely proportional to the square of the distance between the particles.

$$|F_e| = k \cdot \frac{|q_1 q_2|}{r^2}$$

Where k is the Coulomb constant ($k = 8.99 \cdot 10^9 N \cdot m^2/C^2$), q_1, q_2 are the charges of the particles acting an electric force on each other and r is the distance between the particles. The direction of the force depends on the sign of the charge of the two particles. If they have opposite charge signs (one positive and one negative) then they attract each other meaning the direction is towards each other. Otherwise, (both positive or both negative) then they repel each other. From this relationship, the vector form of Coulomb's Law can be written as:

$$\vec{F_e} = k \frac{q_1 q_2}{|r|^3} \vec{r}$$

In my system, the only forces acting are the forces that the dipole acts on the moving particle. Even though in theory the moving particle should also act a force on the particles that make up the dipole, here the dipole is fixed.

To derive the acceleration of the particle, we can use Newton's second law of motion to obtain the following::

$$\vec{F} = m_p \vec{a}_p$$

In a dipole environment, $\vec{F}_{pos}, \vec{F}_{neg}$ are the forces the positive and negative components of the dipole act on the particle, respectively. These forces are calculated by Coulomb's Law as shown above.

$$\vec{a}_p = \frac{\vec{F}_{pos} + \vec{F}_{neg}}{m_p}$$

As the movement is in 3 dimensions we can split this into 6 first order differential equations. We can write the following equations for position and velocity for the x axis:

$$\frac{dx}{dt} = v_x, \frac{dv_x}{dt} = \frac{F_{pos_x} + F_{neg_x}}{m_p}$$

We can write the same for every axis (x, y, z) and receive 6 first order differential equations. Based on this set of first order differential equations I can use numerical methods to model the particle's movement.

3.1.2 Conservation of Energy

Energy is the capacity of a system to perform work. In classical mechanics, a fundamental law states that in a closed system governed by conservative forces (forces where the total work done on a particle depends only on its starting and ending points), the total mechanical energy remains constant over time. In the dipole system, the moving particle possesses two forms of energy: Kinetic Energy (K), related to its

motion, and Electrostatic Potential Energy (U), related to its position within the dipole's field. The formulae for calculating these are:

$$K = \frac{mv^2}{2}, U = \frac{kqQ}{R}$$
$$E_{total} = K + U$$

Where m is the mass, v is the velocity, q is the charge of the particle, Q is the charge of the particle in the surroundings and R is the distance between the particles.

In a closed system where only conservative forces such as the electrostatic force are present, the total mechanical energy remains invariant. This implies that the sum of kinetic and electrostatic potential energy is constant, regardless of the path the particle takes between any two points in time, t_1 and t_2:

$$E_{\text{total}}(t_1) = E_{\text{total}}(t_2)$$

Therefore:

$$\Delta E = E_{\text{total}}(t_1) - E_{\text{total}}(t_2) = 0$$

In my simulation, the change in energy should be as close to 0 as possible in order for the simulation to be physically accurate.

3.2 Numerical Methods

The two numerical methods I will use are Euler's method and Verlet's Velocity integration method.

3.2.1 Mathematical Theory

These numerical methods are based on one of the most important principles in calculus: the differential. The differential describes how a function changes over a very small time step dt. It allows us to estimate the next value of a function using its current value and slope. The mathematical formula for the approximation with the differential is:

$$f(t_1) \approx f(t_0) + dt \cdot f'(t_0)$$

The numerical methods use this formula in order to approximate a solution of a first order differential equation in the form:

$$\frac{dx}{dt} = f(x, t)$$

Meaning that at any given x_0, t_0 we can calculate the derivative at that point. After choosing a step size (dt) we can approximate the next value of the function.

3.2.2 Euler's Method

Euler's numerical method given a differential equation in the above form and a starting x_0, t_0 approximates a list of values $[x_1, x_2, ..., x_n]$ at points $[t_1, t_2, ..., t_n]$ where $t_i = t_0 + i \cdot dt$. This method utilizes the differential to approximate the next step directly, step by step. The main advantage of Euler's method is its simplicity and computational efficiency. However, because it uses only a first order derivative, its accuracy is limited. Below appears the pseudocode of the simple case of Euler's numerical method:

Algorithm 1 Basic Euler's Method

```
procedure EULER(x_0, t_0, Δt, steps, f)
    x ← x_0                                  ▷ Initial value of the dependent variable
    t ← t_0                                  ▷ Initial value of the independent variable
    for i = 1 to steps do
        f_val ← f(x, t)                      ▷ Calculate the slope dx/dt
        x ← x + f_val · Δt                   ▷ Update step: x_{n+1} = x_n + f(x_n, t_n)Δt
        t ← t + Δt                           ▷ Increment time
    end for
    return x
end procedure
```

3.2.3 Verlet's Velocity Method

Verlet's Velocity method is designed for numerically solving second order differential equations, meaning that the acceleration is given as a function. This method is more suitable for physical systems than Euler's method because of its ability to conserve energy of a system. Verlet's method receives a second order differential equation and starting position and velocity. The method updates the position and velocity as shown in the algorithm below:

Algorithm 2 The Velocity Verlet Integration Method

```
procedure VELOCITYVERLET(x_0, v_0, Δt, steps, f)
    x ← x_0
    v ← v_0
    for i = 1 to N do
        a_current ← a(x)                     ▷ Calculate acceleration at current position
        v_half ← v + 0.5 · a_current · Δt    ▷ Velocity half-step
        x ← x + v_half · Δt                  ▷ Full position update
        a_next ← a(x)                        ▷ Calculate acceleration at new position
        v ← v_half + 0.5 · a_next · Δt       ▷ Finish velocity update
    end for
    return x, v
end procedure
```

As can be seen in the above pseudocode, the algorithm updates the velocity firstly by a "half-step", then updates the position based on this "half velocity". After this, the algorithm recalculates the acceleration at the new position and finally updates the velocity to the full step.

3.2.4 Conservation of Energy in Numerical Methods

In order for the approximations to be physically accurate the simulation will aim for the change in energy to be as small as possible. To check this change in energy, I will calculate the energy at the start and end of the particle's movement and hence calculate the change of energy. Later I will compare the two numerical methods and see which one is more accurate according to the principle of conservation of energy.

3.3 Chaos Theory

The scientific study of systems that are highly sensitive to initial conditions is called **Chaos Theory**. While some of these systems are governed by deterministic laws, meaning their future behavior is entirely determined by their initial state with no random elements involved, they appear to behave in a random and unpredictable manner over time. Systems with these properties are called chaotic.

This phenomenon is famously illustrated by the "Butterfly Effect," where a microscopic change in the

starting parameters of a system can lead to vast divergences in the resulting trajectory. An example of a chaotic system in the field of physics is the double pendulum. A pendulum which is connected to another pendulum creates creates highly complex and unpredictable motion.

4 Methodology

4.1 Development Environment and Software Stack

Throughout this project I coded with VSCode and utilized Git with GitHub for version control. The project is in the following repository: https://github.com/t0merez/Particle-Motion-Simulation. I coded with Jupyter Notebooks (.ipynb files) with a Python kernel. I stored all my relevant functions in a separate Python file and imported them into my Jupyter Notebook.

Throughout the coding of the project I used multiple Python libraries for storing data, simulations, including:

NumPy: Used for vectorized 3D math and linear algebra, essential for the acceleration and position updates in the Verlet loop.

Matplotlib: Facilitated the generation of static 3D trajectory plots and distance-over-time graphs.

Plotly Express: Provided interactive 3D visualizations and animations to better observe the particle's behavior.

Pandas: Enabled efficient data management and the precise calculation of energy drift across thousands of simulation steps.

Copy: Used to generate identical particle instances for the divergence tests, ensuring independent trajectory calculations.

4.2 System Configuration

4.2.1 The Electrostatic Dipole Environment

My system includes a dipole (as shown in ***Figure 1***) and a particle in its surroundings in a 3D space. The particle is affected by the electrical force that the two sides of the dipole create. In my Python file I defined the two sides of the dipole as follows:

Table 1: Physical Properties of the Dipole System

Entity	**Position** (x, y, z)	**Charge** (q)	**Mass** (m)
Positive Pole	$(0, 0, 1)$	$+1.0$ C	∞ (Fixed)
Negative Pole	$(0, 0, -1)$	-1.0 C	∞ (Fixed)

4.2.2 Initial Particle Properties

For every simulation, I defined a particle with the following properties: mass, charge, initial position and velocity. I assumed that the particles are released from rest meaning they have no starting acceleration. For all particles, the mass was defined as $1.0kg$ and the charge amplitude as 1.0. Different particles had differing charge signs. I chose the particle's starting position and speed based on physical intuition as to the properties that would make special trajectories.

It should be noted that the physical values used in the simulation (charges of ± 1 C, distances of order 1 m and masses of $1kg$) are not realistic for microscopic particle systems. These values are chosen for numerical convenience, effectively working in normalized units to better visualize the particle dynamics.

4.3 Code Structure and Implementation

4.3.1 Particle Class

The particles have many properties therefore using Object Oriented Programming in Python I created a class called Particle which will hold all the data needed about a certain particle.

Listing 1 Particle Class Definition.

```python
class Particle:
    def __init__(self, position, velocity, mass, charge):
        self.position = np.array(position)
        self.velocity = np.array(velocity)
        self.mass = mass
        self.charge = charge
        self.acceleration = np.zeros(3)
```

4.3.2 Acceleration Calculation

This function calculates the total electric force acting on the particle by summing the forces from both dipole charges. According to the equations shown in ***(Final Equations for Simulation)*** and with the help of NumPy functions I calculate the vector of acceleration of a particle in the following way:

Listing 2 Acceleration Calculation.

```python
def compute_force(particle):
    r_pos = particle.position - positive_side.position
    r_neg = particle.position - negative_side.position
    # Calculate the magnitude of the forces
    F_pos = K * particle.charge * positive_side.charge / np.linalg.norm(r_pos)**2
    F_neg = K * particle.charge * negative_side.charge / np.linalg.norm(r_neg)**2
    # Calculate the direction of the forces
    F_pos_vector = F_pos * r_pos / np.linalg.norm(r_pos)
    F_neg_vector = F_neg * r_neg / np.linalg.norm(r_neg)

    # Total force is the sum of the forces from both poles
    total_force = F_pos_vector + F_neg_vector

    return total_force

# Calculate the acceleration of the particle based on the total force
def calculate_acceleration(particle):
    total_force = compute_force(particle)
    particle.acceleration = total_force / particle.mass
```

4.3.3 Implementation of Euler's Method

The algorithm at ***(Basic Euler's Method)*** showcases the general idea in Euler's numerical method. However in my simulation there is not one differential equation as is here and not one variable but 2 differential equations with 3 dimensions meaning 6 differential equations. In my code I defined a particle object which stores all the charges properties: mass, charge, position vector and velocity vector. Firstly I calculate the acceleration of the charge then I update the next step's velocity based on Euler's method. After that, with the new velocity I update the new position also based on Euler's method. Below is the code:

Listing 3 Euler's Numerical Method for Simulation.

```python
def simulate_Euler(p, dt=1e-7, steps=num_of_steps):
    pos = np.zeros((steps + 1, 3))
    vel = np.zeros((steps + 1, 3))
    # Store initial state
    pos[0] = p.position
    vel[0] = p.velocity
    for i in range(steps):
        calculate_acceleration(p)
        # Update velocity and position using Euler's method
        p.velocity = p.velocity + p.acceleration * dt
        p.position = p.position + p.velocity * dt

        pos[i+1] = p.position
        vel[i+1] = p.velocity

    return pos, vel
```

As can be seen my method returns two lists of the position and the velocity for every step because in the end I will want to do a simulation of the movement not just see where the particle ended up.

4.3.4 Implementation of Velocity Verlet Integration Method

My implementation of Velocity Verlet's method is nearly identical to the algorithm shown in ***(Velocity Verlet Integration Method)*** but I store the positions and velocities in a list instead of just returning the final position and velocity.

Listing 4 Velocity Verlet's Method for Simulation.

```python
def simulate_Verlet(p, dt=1e-7, steps = num_of_steps):
    pos = np.zeros((steps + 1, 3))
    vel = np.zeros((steps + 1, 3))
    # Store initial state
    pos[0] = p.position
    vel[0] = p.velocity
    for i in range(steps):
        calculate_acceleration(p)
        a_current = p.acceleration
        # Update velocity by half-step
        v_half = p.velocity + 0.5 * dt * a_current
        # Update position by full-step
        p.position = p.position + dt * v_half
        # Get new acceleration at the new position
        calculate_acceleration(p)
        a_new = p.acceleration
        # Finish velocity update
        p.velocity = v_half + 0.5 * dt * a_new
        pos[i+1] = p.position
        vel[i+1] = p.velocity
    return pos, vel
```

4.3.5 Parameter Choices

For the simulation of the movements using the numerical methods shown above the value of the size of the step, dt, is very important. If I chose too large of a dt value, I would receive that the particle will jump too much each step and then the movement will not be accurate. On the other hand, if I chose a very small dt value the computation time to receive a simulation that portrays interesting trajectories will be very large. Therefore, in order to keep a trade-off of accuracy and computation time I chose $dt = 10^{-7}$. This value keeps the trajectory accurate and also does not take too much time to compute. Combined with $25,000$ steps the simulation calculates 25 microseconds (0.000025 seconds) of the particle's trajectory with great accuracy and very quickly. Even though this 25 microseconds sounds very little it is enough to clearly see the particle's trajectories as the particle has very high velocity and acceleration due to the strong electric forces used in the simulation..

4.4 Simulating Trajectories of Particles

4.4.1 Simulations of Cases

I defined 4 different particles with varying starting conditions and simulated their movements using Verlet's method for reasons explained in Section 6. I defined particles which have the following trajectories relative to the dipole:

Stable Orbit: A particle which orbits around the dipole but not on the same axis all the time.

Figure 8: A particle that moves like a figure 8 around the dipole.

Scattering Particle: A particle that moves in an unpredictable manner and scatters around the dipole.

Circular Particle: A particle that moves in a perfect circular motion around the dipole. It does roughly two thirds of a circle in its motion.

4.4.2 Comparison of Energy Conservation between Numerical Methods

For each trajectory simulation described above I calculated the change in energy between the start and end points. I did this calculation for each numerical method and compared.

4.4.3 Investigating Chaotic Properties of the System

For multiple cases, I defined two particles which differ only in their starting position or starting velocity but only by very small values. For these two particles which differ slightly I plotted a log-distance over time graph and also plotted their trajectories on the same 3D graph. With the help of these two graphs I can conclude if the system is chaotic.

5 Results

5.1 Trajectory Simulations

Below I will describe the particles I defined and present their interesting trajectories. All of these trajectories were simulated with Verlet's integration numerical method. For all simulations the positive side of the dipole is red and the negative side is the blue. In addition, the starting point is green and the ending point is pink where the blue line is the trajectory that connects them.

5.1.1 Stable Orbit

This is the case of a negatively charged particle in a static dipole field. The particle exhibits a complex but bounded orbital pattern, oscillating between the positive (red) and negative (blue) poles. This motion occurs because the particle is alternately attracted and repelled by the two charges of the dipole.

Table 2: Initial State Configuration for Particle Trajectory

Property	**Initial Value**
Position $\mathbf{r}_0$	$(1.5, 0.0, 0.5)$
Velocity $\mathbf{v}_0$	$(0.0, 2.0, 0.0)$
Charge q	-1.0 C
Mass m	1.0 kg

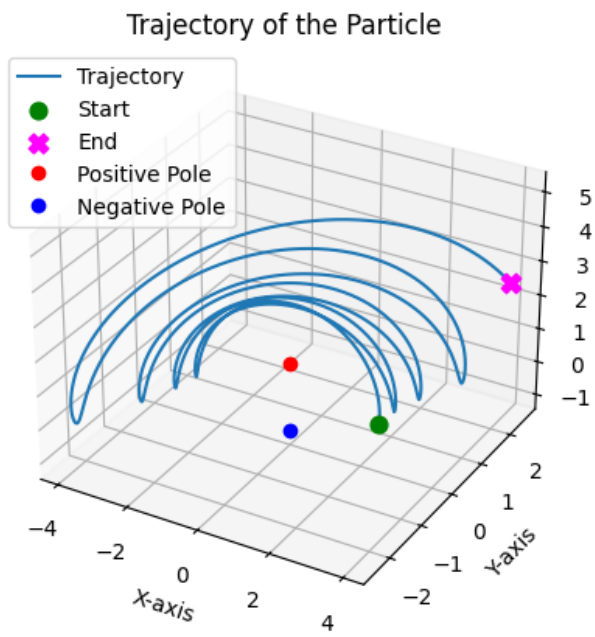


Figure 2

5.1.2 Sling Shot

This case demonstrates motion in a region where the electric force changes very rapidly. The particle enters a tight orbital loop near the dipole center before being ejected from the dipole.

Table 3: Initial State Configuration for Case 2

Property	**Initial Value**
Position $\mathbf{r}_0$	$(0.5, 0.0, 0.0)$
Velocity $\mathbf{v}_0$	$(0.1, 0.0, 0.1)$
Charge q	-1.0 C
Mass m	1.0 kg

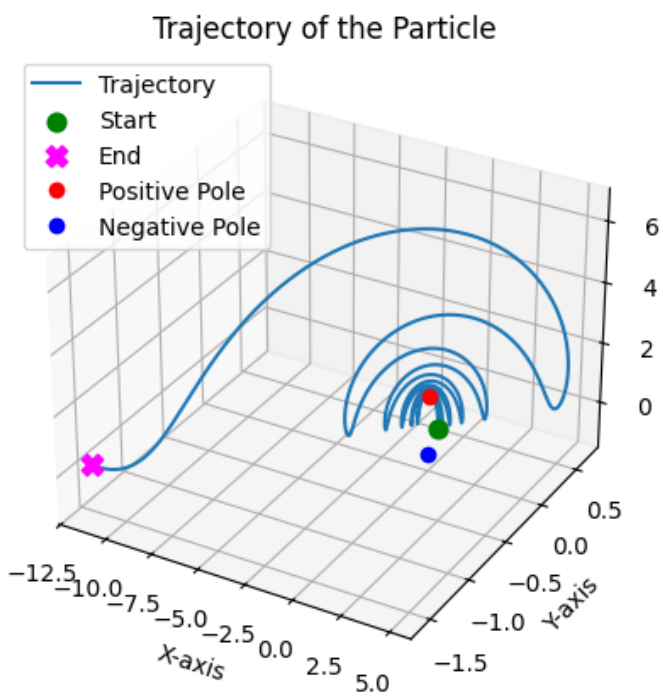


Figure 3

5.1.3 Scattering Movement

In this case the particle exhibits a complex, non-repeating pattern, oscillating unpredictably between the positive and negative poles.

Table 4: Initial State Configuration for Case 3

Property	**Initial Value**
Position $\mathbf{r}_0$	$(-2.0, 1.0, 1.5)$
Velocity $\mathbf{v}_0$	$(1.2, -0.5, -0.8)$
Charge q	-1.0 C
Mass m	1.0 kg

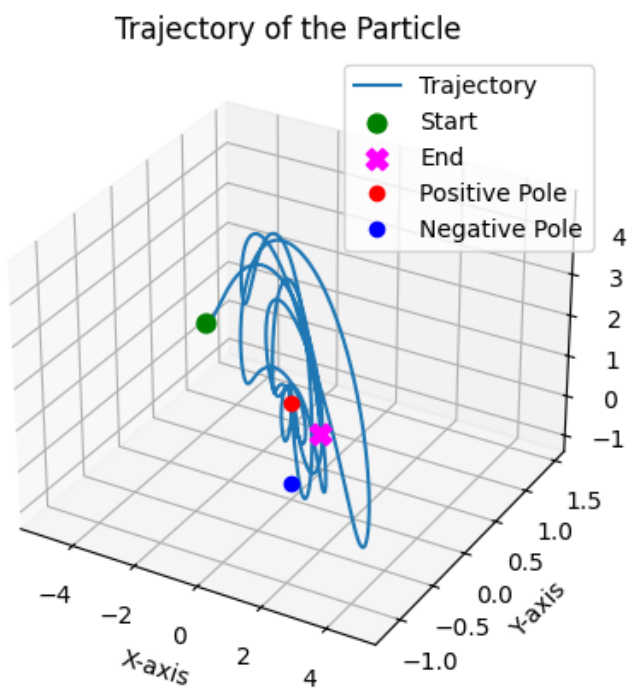


Figure 4

5.1.4 Circular Motion

With no initial Y-velocity, the particle oscillates in the XZ-plane, tracing an arc.

Table 5: Initial State Configuration for Case 4

Property	**Initial Value**
Position $\mathbf{r}_0$	$(2.0, 0.0, -1.0)$
Velocity $\mathbf{v}_0$	$(-1.5, 0.0, 1.5)$
Charge q	-1.0 C
Mass m	1.0 kg

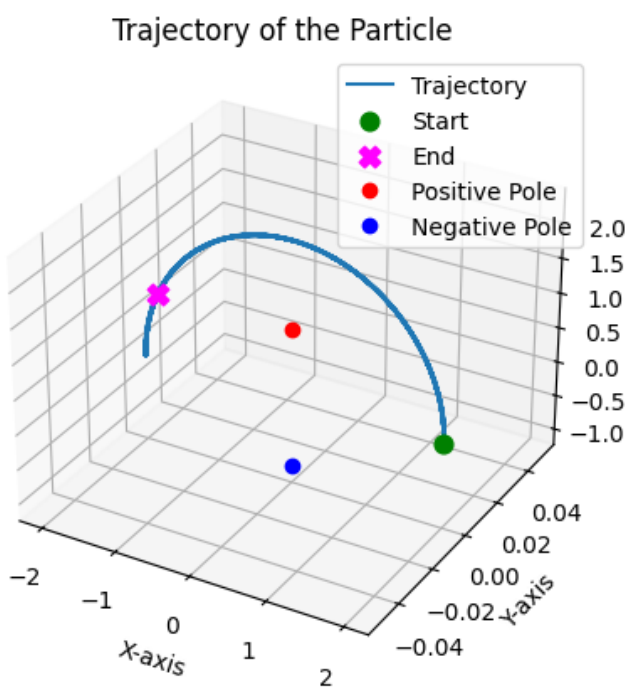


Figure 5

5.2 Energy Conservation of Numerical Methods

For each simulation above I calculated the energy change for both numerical methods and received the following table:

Table 6: Comparison of Energy Conservation (ΔE) between Numerical Methods

Case	**Motion Type**	**Euler ΔE (J)**	**Verlet ΔE (J)**
1	Stable Orbit	-1.4800	-0.0064
2	Sling Shot	-3.1210	-0.0091
3	Scattering Motion	-2.8540	-0.0072
4	Circular Motion	-1.1520	-0.0041

As can be clearly seen in every case the change in energy using Verlet's method is much smaller than Euler's method. This means that Verlet's method approximates energy conservation much better than Euler's method.

5.3 Chaotic Properties

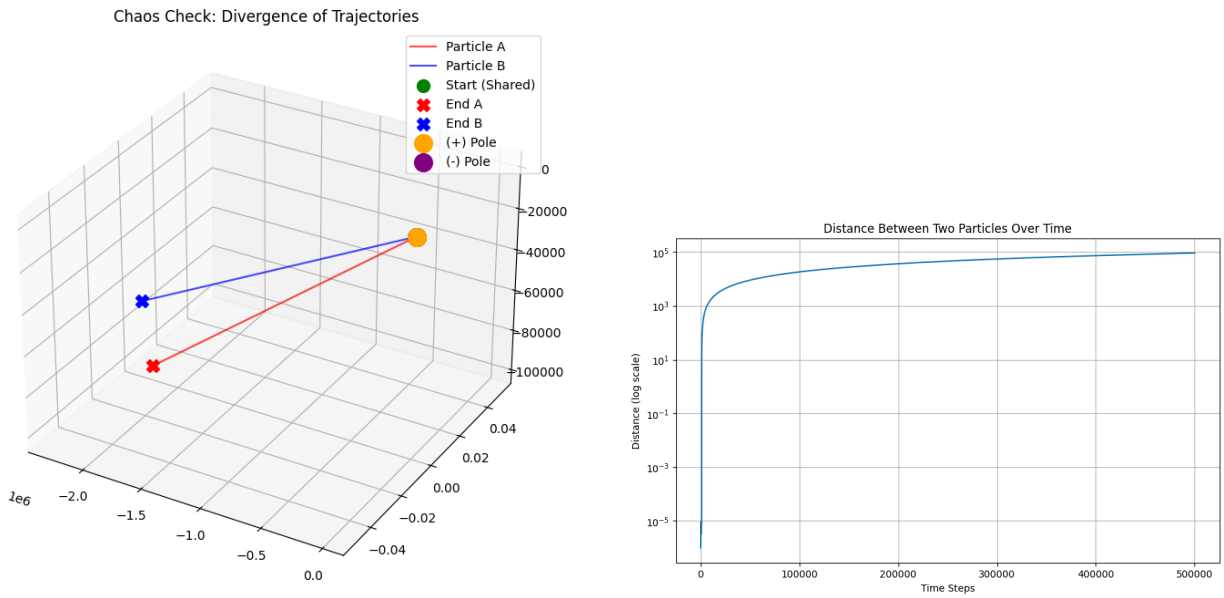
In this section of the project I investigated whether the system is chaotic by comparing two particles with nearly identical initial conditions. I will present two cases which show the chaos of the system with two figures: a log-distance over time graph between two particles with microscopic changes in initial conditions and a simulation of both particles on the same axis.

5.3.1 Change in Initial Position

This case shows the chaotic properties of the system where the only difference between the particles is in the initial position.

Table 7: Initial State Configuration for Initial Change in Position

Property	**Particle 1**	**Particle 2**
Position $\mathbf{r}_0$	$(1.5, 0.0, 0.5)$	$(1.5 + 10^{-6}, 0.0, 0.5)$
Velocity $\mathbf{v}_0$	$(0.0, 0.0, 50.0)$	$(0.0, 0.0, 50.0)$
Charge q	-1.0 C	-1.0 C
Mass m	1.0 kg	1 kg



(a) 3D Divergence of Trajectories

(b) Log-Distance Separation

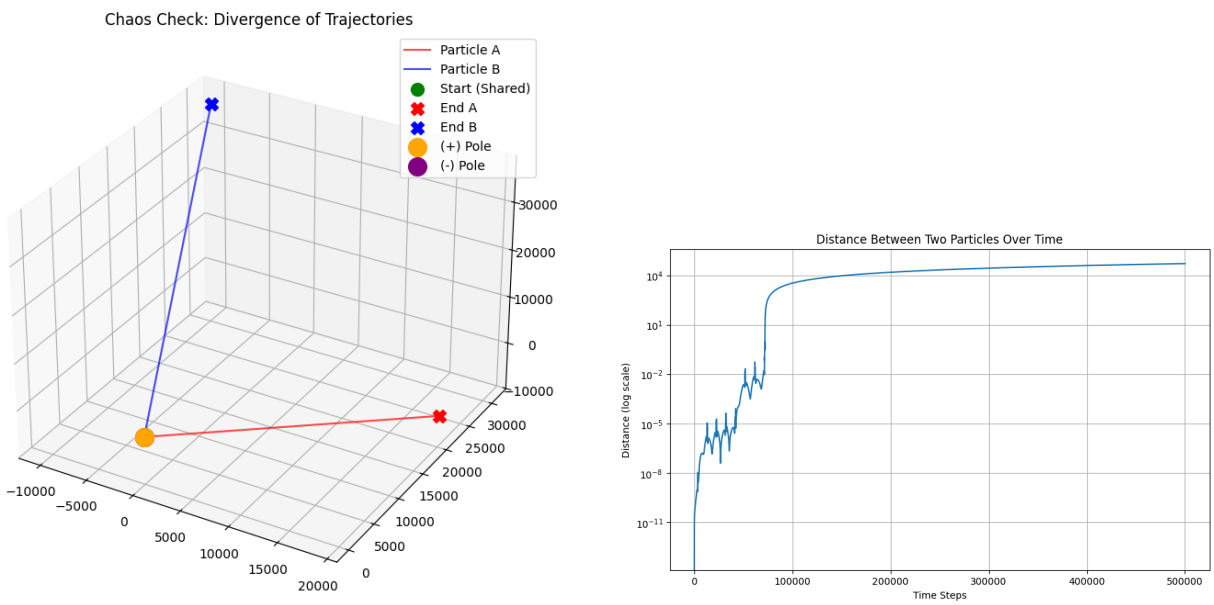
Figure 6: A microscopic position change of 10^{-6} leads to macroscopic separation, confirming deterministic chaos.

5.3.2 Change in Initial Velocity

This case examines sensitivity to changes in initial velocity rather than position.

Table 8: Initial State Configuration for Initial Change in Velocity

Property	**Particle 1**	**Particle 2**
Position $\mathbf{r}_0$	$(-5.0, 1.0, 0.5)$	$(-5.0, 1.0, 0.5)$
Velocity $\mathbf{v}_0$	$(3500.0, 0.0, 0.0)$	$(3500.0 + 10^{-6}, 0.0, 0.0)$
Charge q	-1.0 C	-1.0 C
Mass m	1.0 kg	1 kg



(a) 3D Divergence of Trajectories

(b) Log-Distance Separation

Figure 7: A microscopic velocity change of 10^{-6} leads to macroscopic separation, confirming deterministic chaos.

6 Discussion

The results of this project provide a comprehensive look at the dynamics of a charged particle in a static dipole field. By analyzing four distinct cases and the chaos assessment, several key conclusions can be drawn regarding both the physics of the system and the computational methods used to model it.

A primary goal in this project was to compare Verlet's integration method and Euler's numerical method in physical system models. As shown in **5.2** Verlet's method conserves energy much better than Euler's method. This confirms that Euler's method, while computationally simple, is unsuitable for physical simulations due to its lack of energy conservation. This explains why I used Verlet's integration method for the simulations.

I successfully simulated several diverse trajectories. These simulations demonstrate the powerful utility of numerical methods to understand physical systems based on physical laws. The simulation could produce different trajectories, including a bounded orbit and an open escape path. These trajectories match the expected behavior of particles moving under attractive and repulsive electric forces. Furthermore, the circular motion test (**5.1.4**) acted as a sanity check; the simulation correctly handled the absence of Y-axis momentum, maintaining perfect 2D symmetry.

The results in **5.3** clearly show the chaotic properties of the dipole system. By introducing a microscopic change of 10^{-6} to an initial condition, the system eventually produced a large divergence in the particle's final state. The log-distance plot shows approximately linear growth, indicating exponential divergence between trajectories. This shows that while the physics is governed by fixed, non-random laws, the system is sensitive to initial conditions, resulting in unpredictable movement trajectories which seem random.

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